SIMATS ENGINEERING
CO3- ASSESSMENT TOOL 2
SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING **COURSE CODE: ECA09**

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped-BL3,BL4,BL5**

Topics Covered:
Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling.Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyohase Decomposition -5G Decimation/Interpolation: For mmWave beamforming.

Assessment Tool number	**: CO3-AT2**
Assessment Tool Weightage	**: 35%**
Assessment Tool Name	**: Simulation Task**
Duration of this assessment	**: 60 Minutes**
Marks	**: 25 Marks**
Type of Assessment Conduction	**: Self Learning (Home Task)**

- Each student will be assigned one simulation-based problem related to the course topics.
- Solve the assigned simulation problem independently.
- Develop and run the simulation using MATLAB/Simulink.
- Analyze and present the results with graphs and screenshots.
- Upload the simulation code and output to GitHub.
- Submit the GitHub repository link in Google Classroom (GCR).

12. Bartlett Spectral Estimation

Simulate Bartlett spectral estimation for a signal divided into eight segments and evaluate variance reduction compared to the periodogram method. (25) (BL4) (WK4)

Given Data:

- Signal Length = 4096 samples
- Sampling Frequency = 4 kHz
- Number of Segments = 8

Tasks:

1. Divide the signal into equal segments.
2. Compute periodogram of each segment.
3. Average the spectra.
4. Plot Bartlett estimate.
5. Compare with standard periodogram.
6. Analyze variance reduction.

ANSWER:

Bartlett Spectral Estimation

Aim:

To simulate the Bartlett spectral estimation method by dividing a signal into eight equal segments, computing the periodogram of each segment, averaging the spectra, and comparing the result with the standard periodogram to evaluate variance reduction.

Software Required

- MATLAB R2022a

Given Data,

Parameter	Value
Signal Length	4096 Samples
Sampling Frequency	4000 Hz
Number of Segments	8
Segment Length	512 Samples

Theory

- The Periodogram is one of the simplest methods used to estimate the Power Spectral Density (PSD) of a signal. It computes the spectrum using the complete signal, but the estimate generally has high variance, making the spectrum noisy.

- The Bartlett Spectral Estimation method reduces this variance by dividing the signal into several equal non-overlapping segments. The periodogram of each segment is calculated separately, and the resulting spectra are averaged. Averaging decreases the variance of the estimate and produces a smoother power spectrum. However, because each segment is shorter than the original signal, the frequency resolution is slightly reduced.

- Bartlett's method is widely used in digital signal processing because it provides a better compromise between variance and frequency resolution than the standard periodogram.

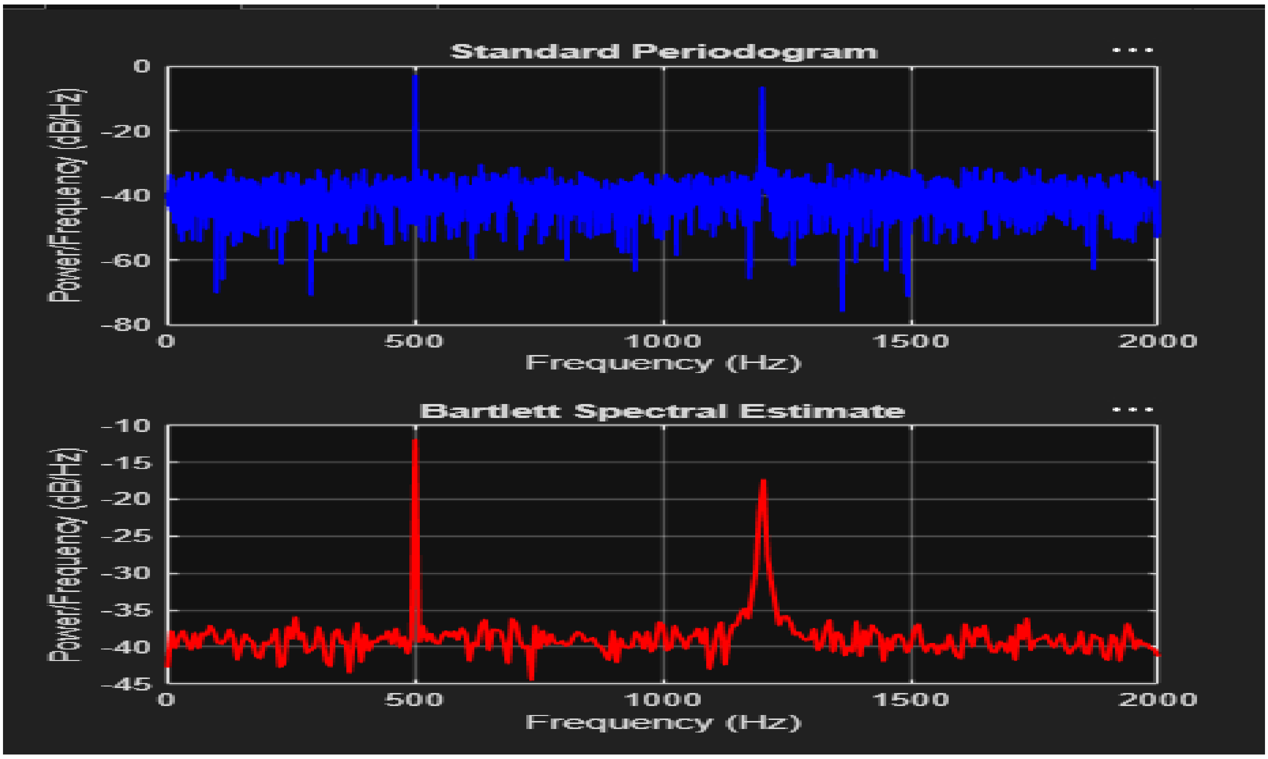
MATLAB Program

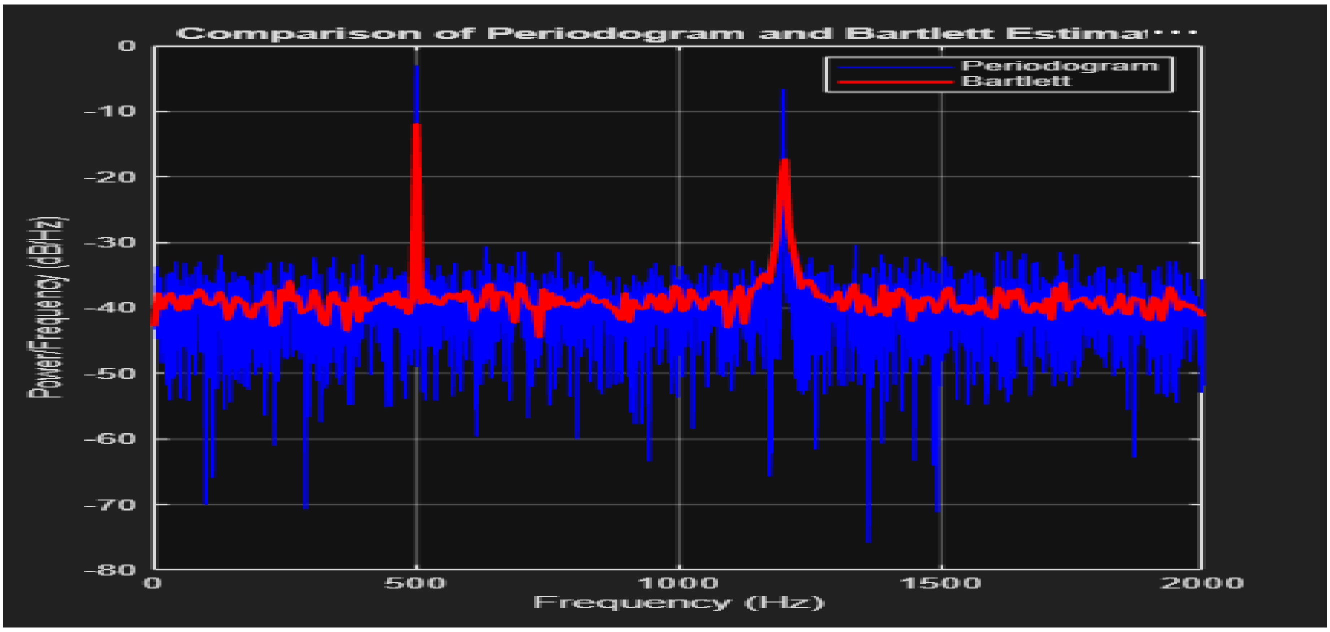
```
clc;
clear;
close all;
%% Parameters
Fs = 4000;          % Sampling Frequency
N = 4096;           % Signal Length
segments = 8;         % Number of Segments
L = N/segments;       % Segment Length
%% Generate Test Signal
t = (0:N-1)/Fs;
x = sin(2*pi*500*t) ...
   +0.7*sin(2*pi*1200*t) ...
   +0.5*randn(1,N);
%% Standard Periodogram
[Pper,f] = periodogram(x,[],N,Fs);
```

```
%% Bartlett Spectral Estimation
BartlettPSD = zeros(L/2+1,1);
for k = 1:segments
    startIndex = (k-1)*L + 1;
    endIndex = k*L;
    xSeg = x(startIndex:endIndex);
    [Pxx,fB] = periodogram(xSeg,[],L,Fs);
    BartlettPSD = BartlettPSD + Pxx;
end
BartlettPSD = BartlettPSD/segments;
%% Plot Results
figure;
subplot(2,1,1)
plot(f,10*log10(Pper),'b','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title('Standard Periodogram');
subplot(2,1,2)
plot(fB,10*log10(BartlettPSD),'r','LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title('Bartlett Spectral Estimate');
%% Comparison Plot
figure;
plot(f,10*log10(Pper),'b');
hold on;
plot(fB,10*log10(BartlettPSD),'r','LineWidth',2);
grid on;
```

```
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
legend('Periodogram','Bartlett');
title('Comparison of Periodogram and Bartlett Estimate');
```

Output





Result Analysis

The Bartlett method successfully reduced the variance of the spectral estimate by averaging the periodograms of eight equal segments. Compared with the standard periodogram, the Bartlett estimate produced a smoother and more stable spectrum while still identifying the major frequency components of the signal. The reduction in variance improves the reliability of the spectral estimate, although the shorter segment length results in a slight loss of frequency resolution.

Conclusion

The Bartlett spectral estimation technique was successfully implemented in MATLAB. The signal was divided into eight equal segments, and the periodograms of each segment were averaged to obtain the final power spectral density estimate. Compared with the standard periodogram, Bartlett's method significantly reduced variance and produced a smoother spectrum while accurately identifying the signal frequency components.